# Continued Fractions

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## 1 History

Continued fractions were often times used off-handedly in the last 2000 years, but the foundations of continued fractions were not established until late 1600's, early 1700's by John Wallis, see a fuller account in the Mathematics Archives [1]. Below is an except from the Mathematics Archives [1].

> In his book Opera Mathematica (1695) Wallis laid some basic groundwork for continued fractions. He explained how to compute the nth convergent and discovered some now familiar properties of convergents. It was also in this work that the term "continued fraction" was first used.
>
> The Dutch mathematician and astronomer Christiaan Huygens (1629-1695) was the first to demonstrate a practical application of continued fractions. He wrote a paper explaining how to use the convergents of a continued fraction to find the best rational approximations for gear ratios. These approximations enabled him to pick the gears with the correct number of teeth. His work was motivated impart by his desire to build a mechanical planetarium.
>
> While the work of Wallis and Huygens began the work on continued fractions, the field of continued fractions began to flourish when Leonard Euler (1707-1783), Johan Heinrich Lambert (1728-1777), and Joseph Louis Lagrange (1736-1813) embraced the topic. Euler laid down much of the modern theory in his work De Fractionlous Continious (1737). He showed that every rational can be expressed as a terminating simple continued fraction. He also provided an expression for e in continued fraction form. He used this expression to show that e and $e^2$ are irrational. He also demonstrated how to go from a series to a continued fraction representation of the series, and conversely.
>
> Lambert generalized Euler's work on e to show that both $e^x$ and $\tan(x)$ are irrational if x is rational. Lagrange used continued fractions to find the value of irrational roots. He also proved that a real root of a quadratic irrational is a periodic continued fraction.
>
> The nineteenth century can probably be described as the golden age of continued fractions. As Claude Brezinski writes in History of Continued Fractions and Padre Approximations, "the nineteenth

century can be said to be popular period for continued fractions." It was a time in which "the subject was known to every mathematician." As a result, there was an explosion of growth within this field. The theory concerning continued fractions was significantly developed, especially that concerning the convergents. Also studied were continued fractions with complex variables as terms. Some of the more prominent mathematicians to make contributions to this field include Jacobi, Perron, Hermite, Gauss, Cauchy, and Stieljes. By the beginning of the 20th century, the discipline had greatly advanced from the initial work of Wallis.

# 2 Definition

**Definition 2.1** (Finite General Continued Fraction)**.** *A **finite general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{\ddots + \cfrac{b_n}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a general continued fraction.

**Definition 2.2** (General Continued Fraction)**.** *A **general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}$$

**Definition 2.3** (Finite Standard Continued Fraction)**.** *A **finite standard** continued fraction is defined as:*

$$[a_0; a_1, a_2, \cdots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a standard continued fraction.

**Definition 2.4** (Standard Continued Fraction)**.** *A **standard** continued fraction is defined as:*

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}$$

*This form is also referred to as the simple continued fraction or regular continued fraction, or the canonical form.*

## 2.1 Rational Examples

Every rational number has a unique representation in the standard continued fraction form. Moreover, every rational number can be represented using only the finite continued fraction form. Below are a few examples.

$$\frac{1}{8} = [0;8] = \mathbf{0} + \frac{1}{\mathbf{8}}$$

$$\frac{2}{5} = [0;2,2] = \mathbf{0} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{163}{7} = [23;3,2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{-13}{5} = [-3;2,2] = \mathbf{-3} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

## 2.2 Generation of continued fractions

The simple continued fraction representation of a number is both easy to verify and easy to generate. Here we go through two common ways of generating the simple continued fraction representation of a number.

### 2.2.1 Rational numbers

Given a rational number $p/q$ we can use Euclid's algorithm for determining the greatest common divisor (GCD) between the numbers $p$ and $q$.
Using the example of $\frac{163}{7}$ as listed above we get the following steps:

See how many times 7 goes into 163 evenly

$$163 = \mathbf{23} \cdot 7 + \mathbf{2}$$

See how many times 2 goes into 7 evenly

$$7 = \mathbf{3} \cdot 2 + \mathbf{1}$$

See how many times 1 goes into 2 evenly

$$2 = \mathbf{2} \cdot 1 + \mathbf{0}$$

Notice that the resulting divisor values exactly match the corresponding simple continued fraction representation of $\frac{163}{7}$ as shown above. In particular

$$\frac{163}{7} = [23;3,2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

### 2.2.2 General real numbers

We can apply the same reasoning as for the rational numbers to general real numbers.
Using $\sqrt{2} = 1.4142135623730951\ldots$ as an example.

Remove the largest integer from value

$$\sqrt{2} = \mathbf{1} + 0.4142135623730951\ldots$$

Take the reciprocal of the result and remove the largest integer from that

$$1/0.4142135623730951\cdots = 2.414213562373095 = \mathbf{2} + .414213562373095$$

Repeat the process

$$1/0.414213562373095\cdots = 2.41421356237309 = \mathbf{2} + .414213562373095$$

These calculations seem to imply that

$$\sqrt{2} = [1; 2, 2, 2, \ldots] = \mathbf{1} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \ddots}}}$$

We will see in a later section that this is the exact continued fraction representation of $\sqrt{2}$.

Moreover, we will show that in general if $\sqrt{n}$ is not an integer then it will always have a periodic repeating infinite continued fraction representation.

The algorithm presented here will work for every number as long as a decimal representation is available for said number.

# 3 Useful lemmas

**Definition 3.1** (Convergents)**.** *A convergent $\frac{p_n}{q_n}$ of a simple continued fraction $[a_0; a_1, a_2, a_3, \ldots]$ is defined as:*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, a_3, \ldots, a_n]$$

*For reasons that will become apparent later we define $p_{-1} = 1, p_{-2} = 0$ and $q_{-1} = 0, q_{-2} = 1$.*

**Lemma 3.2** (Recurrence Relation)**.** *For $\frac{p_n}{q_n}$ the $n^{th}$ convergent of $[a_0; a_1, a_2, a_3, \ldots]$.*

*We can find $p_n, q_n$ using the previous convergents through the following recurrence relation:*

$$\frac{p_n}{q_n} = \frac{a_n \cdot p_{n-1} + p_{n-2}}{a_n \cdot q_{n-1} + q_{n-2}}$$

*Proof.*

$$\frac{p_0}{q_0} = a_0 = \frac{a_0 \cdot 1 + 0}{a_0 \cdot 0 + 1} = \frac{a_0 \cdot p_{-1} + p_{-2}}{a_0 \cdot q_{-1} + q_{-2}}$$
$$\frac{p_1}{q_1} = a_0 + \frac{1}{a_1} = \frac{a_0 a_1 + 1}{a_1} = \frac{a_1 \cdot a_0 + 1}{a_1 \cdot 1 + 0}$$
$$\frac{p_2}{q_2} = a_0 + \frac{1}{a_1 + \frac{1}{a_2}} = a_0 + \frac{a_2}{a_1 a_2 + 1} = \frac{a_0 a_1 a_2 + a_0 + a_2}{a_1 a_2 + 1} = \frac{a_2 \cdot (a_0 a_1 + 1) + a_0}{a_2 \cdot a_1 + 1}$$

Assume true for $k$ and prove for $k+1$:

$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_k + \cfrac{1}{a_{k+1}}}}}$$
$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{\cfrac{a_k a_{k+1} + 1}{a_{k+1}}}}}$$

Let $a_k^{'} = \frac{a_k a_{k+1} + 1}{a_{k+1}}$ and use the assumption for $\frac{p_k}{q_k}$:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{\frac{a_k a_{k+1} + 1}{a_{k+1}} \cdot p_{k-1} + p_{k-2}}{\frac{a_k a_{k+1} + 1}{a_{k+1}} \cdot q_{k-1} + q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{(a_k a_{k+1} + 1) \cdot p_{k-1} + a_{k+1} p_{k-2}}{(a_k a_{k+1} + 1) \cdot q_{k-1} + a_{k+1} q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1}(a_k p_{k-1} + p_{k-2}) + p_{k-1}}{a_{k+1}(a_k q_{k-1} + q_{k-2}) + q_{k-1}}$$

Again using the recursive assumption for $p_k$ and $q_k$ we finish the inductive proof:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1} p_k + p_{k-1}}{a_{k+1} q_k + q_{k-1}}$$

□

**Lemma 3.3** (Determinant Identity)**.** *For $\frac{p_n}{q_n}$, the $n^{th}$ convergent of $[a_0; a_1, a_2, a_3, \ldots]$:*

$$p_{n-1} \cdot q_n - p_n \cdot q_{n-1} = (-1)^n$$

*Proof.* We proceed by verifying the base cases and then using mathematical induction on $n$.

**Base Cases:**

$$n = 0: \quad p_{-1} \cdot q_0 - p_0 \cdot q_{-1} = 1 \cdot 1 - a_0 \cdot 0 = 1 = (-1)^0,$$
$$n = 1: \quad p_0 \cdot q_1 - p_1 \cdot q_0 = a_0 \cdot a_1 - (a_0 a_1 + 1) \cdot 1 = -1 = (-1)^1,$$

**Induction Hypothesis:** Assume that for $k$, the statement holds:

$$p_{k-1} \cdot q_k - p_k \cdot q_{k-1} = (-1)^k.$$

**Inductive Step:** We need to prove the statement for $k+1$:

$$\begin{aligned} p_k \cdot q_{k+1} - p_{k+1} \cdot q_k &= p_k \cdot (a_{k+1}q_k + q_{k-1}) - (a_{k+1}p_k + p_{k-1}) \cdot q_k \\ &= (p_k \cdot q_{k-1} - p_{k-1} \cdot q_k) \\ &= -(p_{k-1} \cdot q_k - p_k \cdot q_{k-1}) = -(-1)^k = (-1)^{k+1}. \end{aligned}$$

Therefore, $p_k \cdot q_{k+1} - p_{k+1} \cdot q_k = (-1)^{k+1}$, completing the induction step. □

**Lemma 3.4** (Difference Identity)**.** *For $\frac{p_n}{q_n}$ the $n^{th}$ convergent of $[a_0; a_1, a_2, a_3, \dots]$.*
*We can define a closed form for the difference of two sequential convergents:*

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n \cdot q_{n-1}}$$

*Proof.*

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{p_{n-1} \cdot q_n - p_n \cdot q_{n-1}}{q_n q_{n-1}}$$

Using Lemma 3.3 we have

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n q_{n-1}}$$

□

# 4 Relation to alternating Series

**Theorem 4.1** (Series to Continued Fraction)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 \cfrac{a_3^2}{\cdots + \cfrac{a_{n-1}^2}{a_n - a_{n-1}}}}}}$$

*Proof.* **Proof by induction**
**Base Case n=1:**

$$\sum_{k=1}^{1} \frac{(-1)^{k-1}}{a_k} = \frac{1}{a_1} = \frac{1}{a_1}$$

**Base Case n=2:**

$$\begin{aligned}\sum_{k=1}^{2}\frac{(-1)^{k-1}}{a_k} &= \frac{1}{a_1}-\frac{1}{a_2}\\ &= \frac{a_2-a_1}{a_1\cdot a_2}\\ &= \frac{1}{\frac{a_1\cdot a_2}{a_2-a_1}}\\ &= \frac{1}{a_1+\frac{a_1^2}{a_2-a_1}}\end{aligned}$$

**Inductive Hypothesis:** Assume the statement is true for $n$:

$$\sum_{k=1}^{n}\frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1+\cfrac{a_1^2}{a_2-a_1+\cfrac{a_2^2}{a_3-a_2\cfrac{a_3^2}{\ddots+\cfrac{a_{n-1}^2}{a_n-a_{n-1}}}}}}$$

**Inductive Step:** Prove for $n+1$.

The idea of the proof is to re-write the summation for $n+1$ as a summation over $n$ terms by obtaining a new $\widehat{a_n}$ term.

$$\begin{aligned}\sum_{k=1}^{n+1}\frac{(-1)^{k-1}}{a_k} =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+\frac{(-1)^{n-1}}{a_n}+\frac{(-1)^n}{a_{n+1}}\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{a_n}-\frac{1}{a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{a_{n+1}-a_n}{a_n\cdot a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\frac{1}{\widehat{a_n}}\end{aligned}$$

Note that

$$\widehat{a_n}=\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}=\frac{(a_{n+1}-a_n)\cdot a_n+a_n^2}{a_{n+1}-a_n}=a_n+\frac{a_n^2}{a_{n+1}-a_n}$$

Hence, plugging in this new $\widehat{a_n}$ into our inductive hypothesis we get:

$$\sum_{k=1}^{n+1} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\widehat{a_n} - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\left(a_n + \frac{a_n^2}{a_{n+1} - a_n}\right) - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{a_n - a_{n-1} + \frac{a_n^2}{a_{n+1} - a_n}}}}}}$$

Which proves the theorem for $n+1$ since this matches the form we were expecting for $n+1$. □

**Theorem 4.2** (Euler's Continued Fraction Formula)**.** *https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula*

$$\sum_{k=0}^{n} a_0 a_1 \cdots a_k = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{1 + a_2 - \cfrac{a_3}{\ddots + 1 + a_{n-1} - \cfrac{a_n}{1 + a_n}}}}}$$

*Proof.* We proceed by induction on $n$.

**Base case** $(n = 0)$**.**

$$\sum_{k=0}^{0} a_0 a_1 \cdots a_k = a_0, \qquad \frac{a_0}{1} = a_0.$$

**Base case** $(n = 1)$**.**

$$\text{LHS} = a_0 + a_0 a_1 = a_0(1 + a_1),$$

$$\text{RHS} = \frac{a_0}{1 - \dfrac{a_1}{1 + a_1}} = \frac{a_0(1 + a_1)}{1 + a_1 - a_1} = a_0(1 + a_1).$$

**Inductive hypothesis.** Assume for some $k \geq 1$ that

$$\sum_{j=0}^{k} a_0 a_1 \cdots a_j = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_k}{1 + a_k}}}}.$$

**Inductive step.** We show the formula holds for $k+1$.

Let

$$S_k := \sum_{j=0}^{k} a_0 a_1 \cdots a_j.$$

Then

$$S_{k+1} = S_k + a_0 a_1 \cdots a_{k+1} = S_k \left(1 + a_{k+1} \frac{a_0 a_1 \cdots a_k}{S_k}\right).$$

From the inductive hypothesis, the ratio

$$\frac{a_0 a_1 \cdots a_k}{S_k}$$

is exactly the value of the *last convergent* of the continued fraction, namely

$$\frac{a_k}{1 + a_k}.$$

Hence

$$S_{k+1} = S_k \left(1 + \frac{a_k a_{k+1}}{1 + a_k}\right).$$

Now observe that appending one more level to the continued fraction replaces the final denominator

$$1 + a_k \quad \text{by} \quad 1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}}.$$

Indeed,

$$1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}} = \frac{(1 + a_k)(1 + a_{k+1}) - a_{k+1}}{1 + a_{k+1}} = \frac{1 + a_k + a_{k+1} + a_k a_{k+1}}{1 + a_{k+1}}.$$

This produces exactly the multiplicative factor

$$1 + \frac{a_k a_{k+1}}{1 + a_k}$$

in the numerator, while preserving the continued fraction structure above it.

Therefore,

$$S_{k+1} = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_{k+1}}{1 + a_{k+1}}}}}.$$

This completes the induction. □

**Theorem 4.3** (Infinite Product Alternatives)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \cfrac{a_3}{\ddots + a_{n-1} - 1 + \cfrac{a_{n-1}}{a_n - 1}}}}}$$

*Proof.* **Proof by induction**

We proceed by induction on $n$.

**Base Case: For** $n = 1$, the left-hand side of the equation is:

$$\frac{(-1)^{1-1}}{a_1} = \frac{1}{a_1}$$

And the right-hand side for $n = 1$ is:

$$\frac{1}{a_1}$$

Thus, the equation holds for $n = 1$.

**Base Case: For** $n = 2$, the left-hand side of the equation is:

$$\begin{aligned}\frac{1}{a_1} - \frac{1}{a_1 a_2} &= \frac{a_2 - 1}{a_1 a_2} = \frac{1}{\frac{a_1 a_2}{a_2 - 1}} \\ &= \frac{1}{\frac{a_1(a_2-1)+a_1}{a_2-1}} \\ &= \frac{1}{a_1 + \frac{a_1}{a_2 - 1}}\end{aligned}$$

And the right-hand side for $n = 2$ is:

$$\frac{1}{a_1 + \frac{a_1}{a_2 - 1}}$$

Thus, the equation holds for $n = 2$.

**Inductive Step:** Assume that the equation holds for some $n = K$, that is:

$$\sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1}}}}}$$

We want to prove that the equation holds for $n = K + 1$, the left-hand side becomes:

$$\begin{aligned}
\sum_{k=1}^{K+1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} &= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_K} - \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K+1}} \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left( \frac{1}{a_K} - \frac{1}{a_K a_{K+1}} \right) \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left( \frac{a_{K+1} - 1}{a_K a_{K+1}} \right)
\end{aligned}$$

Define $\widehat{a_K} = \frac{a_K a_{K+1}}{a_{K+1}-1}$

$$= \sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_{K-1} \widehat{a_K}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\widehat{a_K} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1}}{a_{K+1}-1} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1} - a_{K+1} + 1}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{(a_K - 1)\cdot(a_{K+1}-1) + a_K}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1 + \frac{a_K}{a_{K+1}-1}}}}}}$$

□

## 4.1 Example $\ln(2)$

**Lemma 4.4** (Taylor Series for ln)**.**

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{k}$$

*Proof.* This can be shown using the Taylor Series for $\ln(x+1)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series.

Note that

$$\frac{\partial^n \big(\ln(x+1)\big)}{\partial x^n}|_{x=0} = (-1)^{n+1} \cdot (n-1)!$$

Thus plugging this into Taylor's equation we get:

$$\begin{aligned}
\ln(1+x) &= \sum_{m=0}^{\infty} \frac{(x-a)^m f^{(m)}(a)}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m (m-1)!}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m}{m}
\end{aligned}$$

□

**Lemma 4.5** (Continued Fraction for $\ln(2)$)**.**

$$\ln(2) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

*Proof.* Plugging in $x = 1$ to the Taylor expansion of $\ln(x+1)$ and using Theorem 4.1 we obtain the result. □

## 4.2 Example $\frac{\pi}{4}$

**Lemma 4.6** (Taylor Series for arctan)**.**

$$\arctan(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^{2k-1}}{2k-1}$$

*Proof.* This can be shown using the Taylor Series for $\arctan(x)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series. □

**Lemma 4.7** (Continued Fraction for $\pi/4$)**.**

$$\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

*Proof.* Plugging in $x = 1$ to the Taylor expansion of $\arctan(x) = \arctan(1) = \frac{\pi}{4}$ and using Theorem 4.1 we obtain the result. □

# 5 Square root numbers

**Theorem 5.1** (Square Root CF)**.** *For any $x > 0$:*

$$\sqrt{x} = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

*Proof.* Let $y$ denote the value of the continued fraction:

$$y = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

Since the continued fraction is infinite and periodic, we can observe that the denominator starting from the first 2 is equal to $1 + y$. That is:

$$2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}} = 1 + y$$

Therefore, we can rewrite $y$ as:

$$y = 1 + \frac{x-1}{1+y}$$

Multiplying both sides by $(1 + y)$:

$$\begin{aligned} y(1+y) &= (1+y) + (x-1) \\ y + y^2 &= 1 + y + x - 1 \\ y^2 &= x \end{aligned}$$

Since $y$ represents the value of a continued fraction with positive terms and $x > 0$, we have $y > 0$. Therefore:

$$y = \sqrt{x}$$

This proves that the continued fraction equals $\sqrt{x}$. □

# 6 Approximation of rational numbers

A central problem in Diophantine approximation is to determine how closely a real irrational number $x$ can be approximated by rational numbers $p/q$ relative to the size of the denominator $q$. The convergents of the standard continued fraction expansion of $x$ provide the sequence of *best rational approximations*.

## 6.1 Best Rational Approximations

**Definition 6.1** (Best Rational Approximation)**.** *Let $x \in \mathbb{R}$ be an irrational number, and let $p/q$ with $q \geq 1$ and $\gcd(p, q) = 1$ be a rational number.*

*(i) $p/q$ is called a **best rational approximation of the first kind** to $x$ if:*

$$\left| x - \frac{p}{q} \right| < \left| x - \frac{a}{b} \right|$$

*for all rational numbers $a/b \neq p/q$ with $1 \leq b \leq q$.*

*(ii) $p/q$ is called a **best rational approximation of the second kind** to $x$ if:*

$$|qx - p| < |bx - a|$$

*for all rational numbers $a/b \neq p/q$ with $1 \leq b \leq q$.*

**Remark 6.1.** *Every best approximation of the second kind is automatically a best approximation of the first kind, since:*

$$\left| x - \frac{p}{q} \right| = \frac{|qx - p|}{q} \leq \frac{|bx - a|}{q} \leq \frac{|bx - a|}{b} = \left| x - \frac{a}{b} \right|$$

*whenever $1 \leq b \leq q$.*

## 6.2 Confinement and Error Bounds of Convergents

**Lemma 6.2** (Alternating Confinement of Convergents)**.** *Let $x = [a_0; a_1, a_2, \dots]$ be an irrational number, and let $p_n/q_n$ denote its convergents. Then the even convergents form a strictly increasing sequence, the odd convergents form a strictly decreasing sequence, and each even convergent is strictly less than $x$, while each odd convergent is strictly greater than $x$:*

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \frac{p_4}{q_4} < \cdots < x < \cdots < \frac{p_5}{q_5} < \frac{p_3}{q_3} < \frac{p_1}{q_1}$$

*Proof.* Consider the difference between convergents whose indices differ by 2. Using the difference identity $\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n q_{n-1}}$ from Lemma 3.4:

$$\begin{aligned} \frac{p_n}{q_n} - \frac{p_{n-2}}{q_{n-2}} &= \left( \frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right) + \left( \frac{p_{n-1}}{q_{n-1}} - \frac{p_{n-2}}{q_{n-2}} \right) \\ &= \frac{(-1)^{n-1}}{q_n q_{n-1}} + \frac{(-1)^{n-1}}{q_{n-1} q_{n-2}} \\ &= \frac{(-1)^{n-1}(q_n - q_{n-2})}{q_n q_{n-1} q_{n-2}} \end{aligned}$$

By the recurrence relation $q_n = a_n q_{n-1} + q_{n-2}$ from Lemma 3.2, we have $q_n - q_{n-2} = a_n q_{n-1}$. Substituting this gives:

$$\frac{p_n}{q_n} - \frac{p_{n-2}}{q_{n-2}} = \frac{(-1)^{n-1} a_n}{q_n q_{n-2}}$$

Since $a_n \geq 1$ and all denominators $q_k > 0$, the sign of the difference is governed entirely by $(-1)^{n-1}$:

- For $n = 2k$ even: $(-1)^{2k-1} = -1 \implies \frac{p_{2k}}{q_{2k}} > \frac{p_{2k-2}}{q_{2k-2}}$, so the even convergents strictly increase.
- For $n = 2k+1$ odd: $(-1)^{2k} = 1 \implies \frac{p_{2k+1}}{q_{2k+1}} < \frac{p_{2k-1}}{q_{2k-1}}$, so the odd convergents strictly decrease.

Moreover, by Lemma 3.4, $\frac{p_{2k+1}}{q_{2k+1}} - \frac{p_{2k}}{q_{2k}} = \frac{1}{q_{2k+1}q_{2k}} > 0$, so every odd convergent exceeds the preceding even convergent. Since $x = \lim_{n\to\infty} \frac{p_n}{q_n}$, $x$ lies strictly between any two consecutive convergents $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$, completing the proof. □

**Theorem 6.3** (Fundamental Error Bounds)**.** *For any convergent $p_n/q_n$ $(n \geq 0)$ of an irrational number $x$:*

$$\frac{1}{q_n(q_{n+1}+q_n)} < \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}$$

*Proof.* Let $\alpha_{n+1} = [a_{n+1}; a_{n+2}, a_{n+3}, \dots]$ denote the $(n+1)^{\text{th}}$ complete quotient of $x$, so that:

$$x = [a_0; a_1, \dots, a_n, \alpha_{n+1}] = \frac{\alpha_{n+1}p_n + p_{n-1}}{\alpha_{n+1}q_n + q_{n-1}}$$

Subtracting $\frac{p_n}{q_n}$ from both sides:

$$\begin{aligned} x - \frac{p_n}{q_n} &= \frac{\alpha_{n+1}p_n + p_{n-1}}{\alpha_{n+1}q_n + q_{n-1}} - \frac{p_n}{q_n} \\ &= \frac{q_n(\alpha_{n+1}p_n + p_{n-1}) - p_n(\alpha_{n+1}q_n + q_{n-1})}{q_n(\alpha_{n+1}q_n + q_{n-1})} \\ &= \frac{p_{n-1}q_n - p_n q_{n-1}}{q_n(\alpha_{n+1}q_n + q_{n-1})} \end{aligned}$$

By the Determinant Identity (Lemma 3.3), $p_{n-1}q_n - p_n q_{n-1} = (-1)^n$. Taking absolute values:

$$\left| x - \frac{p_n}{q_n} \right| = \frac{1}{q_n(\alpha_{n+1}q_n + q_{n-1})} \tag{1}$$

Since $\alpha_{n+1} = a_{n+1} + \frac{1}{\alpha_{n+2}}$ and $\alpha_{n+2} > 1$, we have:

$$a_{n+1} < \alpha_{n+1} < a_{n+1} + 1$$

Substituting the lower bound $\alpha_{n+1} > a_{n+1}$ into (1) and using $q_{n+1} = a_{n+1}q_n + q_{n-1}$:

$$\alpha_{n+1}q_n + q_{n-1} > a_{n+1}q_n + q_{n-1} = q_{n+1}$$

which yields the upper bound:

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

Since $q_{n+1} \geq q_n$, we also have $\frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}$.

Similarly, substituting the upper bound $\alpha_{n+1} < a_{n+1} + 1$ into (1):

$$\alpha_{n+1}q_n + q_{n-1} < (a_{n+1}+1)q_n + q_{n-1} = q_{n+1} + q_n$$

which establishes the lower bound:

$$\left| x - \frac{p_n}{q_n} \right| > \frac{1}{q_n(q_{n+1}+q_n)}$$

□

## 6.3 Legendre's Approximation Theorem

A famous converse question asks: if a rational number $p/q$ is exceptionally close to $x$, must it be a convergent? Adrien-Marie Legendre answered this affirmatively.

**Theorem 6.4** (Legendre's Approximation Theorem)**.** *Let $x \in \mathbb{R}$. If $p, q$ are integers with $q \geq 1$ such that:*

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}$$

*then $\frac{p}{q}$ is a convergent of the continued fraction expansion of $x$.*

*Proof.* Write:

$$x - \frac{p}{q} = \frac{\epsilon\theta}{q^2}$$

where $\epsilon = \pm 1$ and $0 < \theta < \frac{1}{2}$. Expand the rational number $p/q$ into a finite continued fraction $[a_0; a_1, \ldots, a_n]$. Since any finite continued fraction can be expressed with either an odd or an even number of terms (via $[a_0; \ldots, a_n] = [a_0; \ldots, a_n - 1, 1]$ if $a_n > 1$), we may choose the parity of $n$ so that:

$$(-1)^{n-1} = -\epsilon \quad \Longleftrightarrow \quad (-1)^n = \epsilon$$

Let $p_n/q_n = p/q$, and let $p_{n-1}/q_{n-1}$ be the $(n-1)^{\text{th}}$ convergent of $p/q$. Define $\alpha \in \mathbb{R}$ by:

$$x = \frac{\alpha p_n + p_{n-1}}{\alpha q_n + q_{n-1}}$$

Solving for $\alpha$:

$$\alpha = \frac{p_{n-1} - x q_{n-1}}{x q_n - p_n} = \frac{-(q_{n-1}x - p_{n-1})}{q_n x - p_n}$$

From $x - \frac{p_n}{q_n} = \frac{\epsilon\theta}{q_n^2}$, the denominator is:

$$q_n x - p_n = \frac{\epsilon\theta}{q_n}$$

For the numerator, write:

$$\begin{aligned} x - \frac{p_{n-1}}{q_{n-1}} &= \left( x - \frac{p_n}{q_n} \right) + \left( \frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right) \\ &= \frac{\epsilon\theta}{q_n^2} + \frac{(-1)^{n-1}}{q_n q_{n-1}} \\ &= \frac{\epsilon\theta}{q_n^2} - \frac{\epsilon}{q_n q_{n-1}} \\ &= -\frac{\epsilon}{q_n q_{n-1}} \left( 1 - \frac{\theta q_{n-1}}{q_n} \right) \end{aligned}$$

where we used $(-1)^{n-1} = -\epsilon$. Multiplying by $q_{n-1}$:

$$q_{n-1}x - p_{n-1} = -\frac{\epsilon}{q_n} \left( 1 - \frac{\theta q_{n-1}}{q_n} \right)$$

Substituting both into the expression for $\alpha$:

$$\alpha = \frac{\frac{\epsilon}{q_n}\left(1 - \frac{\theta q_{n-1}}{q_n}\right)}{\frac{\epsilon\theta}{q_n}} = \frac{1 - \frac{\theta q_{n-1}}{q_n}}{\theta} = \frac{1}{\theta} - \frac{q_{n-1}}{q_n}$$

Since $0 < \theta < \frac{1}{2}$, we have $\frac{1}{\theta} > 2$. Furthermore, $0 \leq q_{n-1} \leq q_n$, so $0 \leq \frac{q_{n-1}}{q_n} \leq 1$. Therefore:

$$\alpha = \frac{1}{\theta} - \frac{q_{n-1}}{q_n} > 2 - 1 = 1$$

Since $\alpha > 1$, it has a continued fraction expansion with initial integer part $\lfloor \alpha \rfloor \geq 1$. It follows that:

$$x = [a_0; a_1, \ldots, a_n, \alpha] = [a_0; a_1, \ldots, a_n, b_0, b_1, \ldots]$$

is the continued fraction expansion of $x$. Thus $p/q = p_n/q_n$ is indeed a convergent of $x$. □

## 6.4 Examples and Applications

### 6.4.1 Approximations of $\pi$

The continued fraction expansion of $\pi$ begins:

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, \ldots]$$

Computing the first several convergents:

$$\begin{aligned}
&\frac{p_0}{q_0} = 3 = 3.000000, && |\pi - 3| \approx 0.141593 \\
&\frac{p_1}{q_1} = 3 + \frac{1}{7} = \frac{22}{7} \approx 3.142857, && \left|\pi - \frac{22}{7}\right| \approx 0.001264 < \frac{1}{7^2} \approx 0.0204 \\
&\frac{p_2}{q_2} = \frac{333}{106} \approx 3.141509, && \left|\pi - \frac{333}{106}\right| \approx 8.32 \times 10^{-5} \\
&\frac{p_3}{q_3} = \frac{355}{113} \approx 3.14159292, && \left|\pi - \frac{355}{113}\right| \approx 2.67 \times 10^{-7}
\end{aligned}$$

The third convergent, $\frac{355}{113}$ (discovered by the Chinese astronomer Zu Chongzhi in the 5th century and known as *Milü*), achieves an extraordinary level of precision for a denominator of only 113. The theoretical reason is evident from Theorem 6.3: the next partial quotient $a_4 = 292$ is remarkably large, giving:

$$q_4 = a_4 q_3 + q_2 = 292 \times 113 + 106 = 33102$$

Hence the error is bounded by:

$$\left|\pi - \frac{355}{113}\right| < \frac{1}{q_3 q_4} = \frac{1}{113 \times 33102} \approx 2.673 \times 10^{-7}$$

### 6.4.2 The Golden Ratio: The Most Irrational Number

The golden ratio $\phi = \frac{1+\sqrt{5}}{2}$ has the continued fraction expansion:

$$\phi = [1; 1, 1, 1, \dots]$$

Its convergents are ratios of consecutive Fibonacci numbers $\frac{F_{n+1}}{F_n}$:

$$1, \quad 2, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad \frac{8}{5}, \quad \frac{13}{8}, \quad \frac{21}{13}, \quad \dots$$

Because every partial quotient attains the minimal possible positive integer value $a_k = 1$, the denominators grow at the slowest possible rate ($q_{n+1} = q_n + q_{n-1}$), and the approximation error $\left|\phi - \frac{p_n}{q_n}\right|$ converges to 0 slower than for any other irrational number. For this reason, $\phi$ is often described as the "most irrational" real number.

# 7 Relation to Pell's Equation

Pell's equation is one of the oldest and most celebrated Diophantine equations in number theory. Let $n$ be a positive integer that is not a perfect square. The equations:

$$x^2 - ny^2 = 1 \tag{2}$$
$$x^2 - ny^2 = -1 \tag{3}$$

are known as the **Pell equation** and **negative Pell equation**, respectively. Continued fractions provide the complete and constructive algorithmic solution to both equations.

## 7.1 Convergents and Solutions

The fundamental connection between continued fractions and Pell's equation is that every non-trivial integer solution must be a convergent of $\sqrt{n}$.

**Theorem 7.1** (Pell's Equation Relation)**.** *If $p, q$ are positive integers such that $p^2 - nq^2 = \pm 1$, then $p/q$ is a convergent of $\sqrt{n}$.*

*Proof.* Suppose $p, q \in \mathbb{N}$ satisfy $p^2 - nq^2 = \pm 1$. Factoring the left-hand side:

$$(p - q\sqrt{n})(p + q\sqrt{n}) = \pm 1 \implies |p - q\sqrt{n}| = \frac{1}{p + q\sqrt{n}}$$

Dividing by $q$:

$$\left|\sqrt{n} - \frac{p}{q}\right| = \frac{1}{q(p + q\sqrt{n})}$$

Since $n \geq 2$ is not a square, we have $\sqrt{n} > 1$. If $p^2 - nq^2 = 1$, then $p^2 = nq^2 + 1 > nq^2 \geq 2q^2$, which implies $p > \sqrt{2}q > q$. If $p^2 - nq^2 = -1$, then $p^2 = nq^2 - 1 \geq 2q^2 - 1 \geq q^2$ (since $q \geq 1$), which implies $p \geq q$. In either case:

$$p + q\sqrt{n} > q + q\sqrt{2} > 2q$$

Therefore:

$$\left|\sqrt{n} - \frac{p}{q}\right| = \frac{1}{q(p + q\sqrt{n})} < \frac{1}{2q^2}$$

By Legendre's Approximation Theorem (Theorem 6.4), $\frac{p}{q}$ must be a convergent of the continued fraction expansion of $\sqrt{n}$. □

## 7.2 The Periodic Continued Fraction of $\sqrt{n}$

As shown in Section 5, the continued fraction expansion of $\sqrt{n}$ is periodic. In standard canonical form:

$$\sqrt{n} = [a_0; \overline{a_1, a_2, \ldots, a_{r-1}, 2a_0}]$$

where $a_0 = \lfloor \sqrt{n} \rfloor$ and $r$ is the length of the minimal period. The partial quotients can be computed via the standard recurrence:

$$P_0 = 0, \quad Q_0 = 1, \quad a_0 = \lfloor \sqrt{n} \rfloor$$

and for $k \geq 0$:

$$P_{k+1} = a_k Q_k - P_k, \qquad Q_{k+1} = \frac{n - P_{k+1}^2}{Q_k}, \qquad a_{k+1} = \left\lfloor \frac{\sqrt{n} + P_{k+1}}{Q_{k+1}} \right\rfloor$$

The convergents $p_k/q_k$ satisfy the key identity:

$$p_k^2 - n q_k^2 = (-1)^{k-1} Q_{k+1} \tag{4}$$

## 7.3 Finding the Fundamental Solution

At the end of the period, when $k = r - 1$, the complete quotient satisfies $Q_r = 1$. Substituting $k = r - 1$ and $Q_r = 1$ into (4):

$$p_{r-1}^2 - n q_{r-1}^2 = (-1)^r$$

This yields two cases depending on the parity of the period length $r$:

1. **Even Period ($r$ is even):** Then $(-1)^r = 1$, so:

   $$p_{r-1}^2 - n q_{r-1}^2 = 1$$

   The **fundamental solution** to $x^2 - ny^2 = 1$ is:

   $$(x_1, y_1) = (p_{r-1}, q_{r-1})$$

   In this case, the negative Pell equation $x^2 - ny^2 = -1$ has no integer solutions.

2. **Odd Period ($r$ is odd):** Then $(-1)^r = -1$, so:

   $$p_{r-1}^2 - n q_{r-1}^2 = -1$$

   Thus $(p_{r-1}, q_{r-1})$ is the fundamental solution to the **negative Pell equation** $x^2 - ny^2 = -1$. To solve the positive equation $x^2 - ny^2 = 1$, we take the convergent at the end of the second period, $k = 2r - 1$. Since $2r$ is even:

   $$p_{2r-1}^2 - n q_{2r-1}^2 = (-1)^{2r} = 1$$

   So the fundamental solution to $x^2 - ny^2 = 1$ is:

   $$(x_1, y_1) = (p_{2r-1}, q_{2r-1})$$

## 7.4 Generating All Solutions

Once the fundamental solution $(x_1, y_1)$ to $x^2 - ny^2 = 1$ is found, all positive integer solutions $(x_k, y_k)$ are generated by powers of the fundamental unit $x_1 + y_1\sqrt{n}$:

$$x_k + y_k\sqrt{n} = (x_1 + y_1\sqrt{n})^k \qquad (k \geq 1)$$

Equivalently, the solutions can be computed recursively:

$$x_{k+1} = x_1 x_k + n y_1 y_k$$
$$y_{k+1} = y_1 x_k + x_1 y_k$$

## 7.5 Worked Examples

### 7.5.1 Example: Even Period ($n = 7$)

Consider $x^2 - 7y^2 = 1$. The continued fraction expansion of $\sqrt{7}$ is:

$$\sqrt{7} = [2; \overline{1, 1, 1, 4}]$$

Here $a_0 = 2, a_1 = 1, a_2 = 1, a_3 = 1, a_4 = 4$, so the period length is $r = 4$ (even). Computing the convergents $p_k/q_k$:

$$k = 0: \quad \frac{p_0}{q_0} = 2, \qquad p_0^2 - 7q_0^2 = 4 - 7 = -3$$
$$k = 1: \quad \frac{p_1}{q_1} = 2 + \frac{1}{1} = 3, \qquad p_1^2 - 7q_1^2 = 9 - 7 = +2$$
$$k = 2: \quad \frac{p_2}{q_2} = \frac{1 \times 3 + 2}{1 \times 1 + 1} = \frac{5}{2}, \qquad p_2^2 - 7q_2^2 = 25 - 28 = -3$$
$$k = 3: \quad \frac{p_3}{q_3} = \frac{1 \times 5 + 3}{1 \times 2 + 1} = \frac{8}{3}, \qquad p_3^2 - 7q_3^2 = 64 - 7(9) = 64 - 63 = 1$$

Since $r = 4$ is even, the fundamental solution occurs at $k = r - 1 = 3$:

$$(x_1, y_1) = (8, 3)$$

The next positive solution is given by:

$$x_2 + y_2\sqrt{7} = (8 + 3\sqrt{7})^2 = 64 + 63 + 48\sqrt{7} = 127 + 48\sqrt{7}$$

Checking: $127^2 - 7(48^2) = 16129 - 7(2304) = 16129 - 16128 = 1$.

### 7.5.2 Example: Odd Period ($n = 13$)

Consider $x^2 - 13y^2 = \pm 1$. The continued fraction expansion of $\sqrt{13}$ is:

$$\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$$

The period length is $r = 5$ (odd). Computing convergents up to $k = r - 1 = 4$:

$$\frac{p_0}{q_0} = \frac{3}{1}, \quad \frac{p_1}{q_1} = \frac{4}{1}, \quad \frac{p_2}{q_2} = \frac{7}{2}, \quad \frac{p_3}{q_3} = \frac{11}{3}, \quad \frac{p_4}{q_4} = \frac{18}{5}$$

Testing $p_4/q_4$:

$$p_4^2 - 13q_4^2 = 18^2 - 13(5^2) = 324 - 325 = -1$$

Thus $(18, 5)$ is the fundamental solution to the negative Pell equation $x^2 - 13y^2 = -1$. Squaring the unit $(18 + 5\sqrt{13})$ yields the fundamental solution to $x^2 - 13y^2 = 1$:

$$(18 + 5\sqrt{13})^2 = 324 + 325 + 180\sqrt{13} = 649 + 180\sqrt{13}$$

So $(x_1, y_1) = (649, 180)$, which corresponds to the convergent $p_9/q_9$ at the end of the second period $(2r - 1 = 9)$. Indeed:

$$649^2 - 13(180^2) = 421201 - 421200 = 1$$

# 8 Solving linear Diophantine Equations

A linear Diophantine equation in two variables has the general form:

$$ax + by = c \tag{5}$$

where $a, b, c \in \mathbb{Z}$ are given integers with $a, b \neq 0$, and we seek all integer solutions $(x, y) \in \mathbb{Z}^2$. Continued fractions provide an elegant, constructive method to find both a particular solution and the full parametric family of solutions.

## 8.1 Solvability Criterion

**Lemma 8.1** (Bézout's Solvability Condition)**.** *Let $a, b, c \in \mathbb{Z}$ with $d = \gcd(a, b)$. The equation $ax + by = c$ has an integer solution if and only if $d$ divides $c$ ($d \mid c$).*

*Proof.* If $(x, y)$ is an integer solution, then $d \mid a$ and $d \mid b$, so $d \mid (ax + by) = c$. Conversely, if $d \mid c$, we divide through by $d$ to obtain:

$$a'x + b'y = c'$$

where $a' = a/d, b' = b/d, c' = c/d$, and $\gcd(a', b') = 1$. By Bézout's identity, there exist integers $u, v$ such that $a'u + b'v = 1$. Multiplying by $c'$ gives $a'(c'u) + b'(c'v) = c'$, yielding the integer solution $(x_0, y_0) = (c'u, c'v)$. □

## 8.2 Constructing Particular Solutions via Continued Fractions

Assume $d \mid c$, and reduce the equation to coprime coefficients:

$$a'x + b'y = c' \qquad \text{with } \gcd(a', b') = 1$$

Assuming without loss of generality that $b' > 0$ and $a' > 0$ (handling signs by setting $x = -x'$ or $y = -y'$ as needed), express the rational number $a'/b'$ as a finite simple continued fraction:

$$\frac{a'}{b'} = [a_0; a_1, a_2, \ldots, a_n]$$

Let $p_k/q_k$ denote the convergents. The final convergent is:

$$\frac{p_n}{q_n} = \frac{a'}{b'}$$

Since $\gcd(p_n, q_n) = 1$ and $\gcd(a', b') = 1$ with $q_n, b' > 0$, we have:

$$p_n = a', \qquad q_n = b'$$

Now consider the penultimate convergent $\frac{p_{n-1}}{q_{n-1}}$. By the Determinant Identity (Lemma 3.3):

$$p_{n-1}q_n - p_n q_{n-1} = (-1)^n$$

Negating both sides:

$$p_n q_{n-1} - p_{n-1}q_n = (-1)^{n-1}$$

Substituting $p_n = a'$ and $q_n = b'$:

$$a'q_{n-1} - b'p_{n-1} = (-1)^{n-1}$$

Multiplying both sides by $(-1)^{n-1}c'$:

$$a' \cdot \left[(-1)^{n-1}c'q_{n-1}\right] + b' \cdot \left[(-1)^n c' p_{n-1}\right] = c'$$

This produces an explicit particular solution:

$$x_0 = (-1)^{n-1}c'q_{n-1} = (-1)^{n-1}\frac{c}{d}q_{n-1}, \qquad y_0 = (-1)^n c' p_{n-1} = (-1)^n \frac{c}{d}p_{n-1} \tag{6}$$

## 8.3 The General Solution

**Theorem 8.2** (General Solution of Linear Diophantine Equation)**.** *Let $a, b, c \in \mathbb{Z}$ with $d = \gcd(a, b) \mid c$. If $(x_0, y_0)$ is any particular integer solution to $ax + by = c$, then the complete set of integer solutions is given by:*

$$x = x_0 + \frac{b}{d}t, \qquad y = y_0 - \frac{a}{d}t \qquad (t \in \mathbb{Z})$$

*Proof.* First, we verify that any pair of this form is indeed a solution:

$$\begin{aligned} ax + by &= a\left(x_0 + \frac{b}{d}t\right) + b\left(y_0 - \frac{a}{d}t\right) \\ &= (ax_0 + by_0) + \left(\frac{ab}{d} - \frac{ab}{d}\right)t \\ &= c + 0 = c \end{aligned}$$

Conversely, let $(x, y)$ be an arbitrary integer solution. Then:

$$ax + by = c = ax_0 + by_0$$

Rearranging:

$$a(x - x_0) = b(y_0 - y)$$

Dividing by $d$:

$$\frac{a}{d}(x - x_0) = \frac{b}{d}(y_0 - y)$$

Since $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$, by Euclid's lemma $\frac{b}{d}$ must divide $(x - x_0)$. Thus there exists an integer $t \in \mathbb{Z}$ such that:

$$x - x_0 = \frac{b}{d}t \implies x = x_0 + \frac{b}{d}t$$

Substituting this back into the equation:

$$\frac{a}{d}\left(\frac{b}{d}t\right) = \frac{b}{d}(y_0 - y) \implies y_0 - y = \frac{a}{d}t \implies y = y_0 - \frac{a}{d}t$$

This proves that every integer solution is of the stated form. □

## 8.4 Worked Examples

### 8.4.1 Example 1: Coprime Coefficients

Solve the Diophantine equation:

$$17x + 5y = 1$$

Here $a = 17, b = 5, c = 1$, and $\gcd(17, 5) = 1$, so solutions exist. We compute the continued fraction expansion of $\frac{17}{5}$:

$$\frac{17}{5} = 3 + \frac{2}{5} = 3 + \frac{1}{2 + \frac{1}{2}} = [3; 2, 2]$$

Here the length is $n = 2$ (even). The convergents are:

$$\frac{p_0}{q_0} = \frac{3}{1}$$
$$\frac{p_1}{q_1} = 3 + \frac{1}{2} = \frac{7}{2}$$
$$\frac{p_2}{q_2} = \frac{2 \times 7 + 3}{2 \times 2 + 1} = \frac{17}{5}$$

The penultimate convergent is $\frac{p_1}{q_1} = \frac{7}{2}$. By the Determinant Identity:

$$p_2 q_1 - p_1 q_2 = 17 \times 2 - 7 \times 5 = 34 - 35 = -1$$

Multiplying by $-1$:

$$17(-2) + 5(7) = 1$$

Hence a particular solution is $(x_0, y_0) = (-2, 7)$. By Theorem 8.2, the general solution is:

$$x = -2 + 5t, \qquad y = 7 - 17t \qquad (t \in \mathbb{Z})$$

### 8.4.2 Example 2: Non-Coprime Coefficients

Solve the Diophantine equation:

$$68x + 20y = 12$$

Here $a = 68, b = 20, c = 12$. The greatest common divisor is $d = \gcd(68, 20) = 4$. Since $4 \mid 12$, solutions exist. Dividing the entire equation by 4:

$$17x + 5y = 3$$

From Example 1, the particular solution to $17x + 5y = 1$ is $(-2, 7)$. Multiplying this solution by $c' = 3$:

$$x_0 = 3 \times (-2) = -6, \qquad y_0 = 3 \times 7 = 21$$

Checking: $68(-6) + 20(21) = -408 + 420 = 12$. By Theorem 8.2, the complete family of solutions is:

$$x = -6 + 5t, \qquad y = 21 - 17t \qquad (t \in \mathbb{Z})$$

# 9 Gauss's continued fraction

In 1813, Carl Friedrich Gauss published a celebrated paper [2] introducing what is now known as **Gauss's continued fraction**. Gauss established that the ratio of two contiguous hypergeometric functions can be expressed as an infinite continued fraction. Because the hypergeometric function encompasses a remarkably vast array of elementary and transcendental functions as special or limiting cases, Gauss's continued fraction serves as a master generating machine for continued fraction representations of logarithms, trigonometric and inverse trigonometric functions, exponentials, and powers.

## 9.1 Hypergeometric Functions and Contiguous Relations

**Definition 9.1** (Gaussian Hypergeometric Function)**.** *For complex parameters $a, b, c$ with $c \notin \{0, -1, -2, \dots\}$, the* ***Gaussian hypergeometric function*** ${}_2F_1(a, b; c; z)$ *is defined for $|z| < 1$ by the power series:*

$$ {}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!} $$

*where $(q)_n$ denotes the rising factorial (or Pochhammer symbol):*

$$ (q)_0 = 1, \qquad (q)_n = q(q+1)(q+2)\cdots(q+n-1) = \frac{\Gamma(q+n)}{\Gamma(q)} \quad (n \geq 1) $$

*For $|z| \geq 1$, the function is defined by analytic continuation along the cut plane $\mathbb{C} \setminus [1, \infty)$.*

Two hypergeometric functions are said to be **contiguous** if their parameters differ by integers of magnitude $\pm 1$. Gauss discovered that any hypergeometric function can be linearly expressed in terms of any two functions contiguous to it. To construct the continued fraction, two fundamental contiguous relations are required.

**Lemma 9.2** (Gauss's Contiguous Relations)**.** *Let $f(a, b; c; z) = {}_2F_1(a, b; c; z)$. Then the following two recurrence identities hold:*

$$ f(a, b; c; z) - f(a, b+1; c+1; z) = -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1, b+1; c+2; z) \tag{7} $$

$$ f(a, b+1; c+1; z) - f(a+1, b+1; c+2; z) = -\frac{(b+1)(c-a+1)}{(c+1)(c+2)} z \cdot f(a+1, b+2; c+3; z) \tag{8} $$

*Proof.* We prove identity (7) by comparing the coefficients of $z^n$ on both sides.

Let the series expansion of the left-hand side of (7) be written as $\sum_{n=0}^{\infty} C_n z^n$. For $n = 0$:

$$ C_0 = 1 - 1 = 0 $$

For $n \geq 1$, the coefficient of $z^n$ is:

$$ C_n = \frac{(a)_n (b)_n}{(c)_n n!} - \frac{(a)_n (b+1)_n}{(c+1)_n n!} $$

Factoring out $\frac{(a)_n}{n!}$:

$$C_n = \frac{(a)_n}{n!}\left[\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n}\right]$$

Recall that $(x+1)_n = \frac{x+n}{x}(x)_n$ for any $x \neq 0$. Applying this to $(b+1)_n$ and $(c+1)_n$, we have:

$$(b+1)_n = \frac{b+n}{b}(b)_n, \qquad (c+1)_n = \frac{c+n}{c}(c)_n$$

Substituting these into the bracketed term:

$$\begin{aligned}
\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n} &= \frac{(b)_n}{(c)_n}\left[1 - \frac{c(b+n)}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n}\left[\frac{b(c+n) - c(b+n)}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n}\left[\frac{bc + bn - bc - cn}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n} \cdot \frac{n(b-c)}{b(c+n)} \\
&= -\frac{(c-b)}{c+n} \cdot \frac{n(b)_n}{b(c)_n}
\end{aligned}$$

Notice that:

$$\frac{(b)_n}{b} = (b+1)_{n-1}, \qquad (a)_n = a(a+1)_{n-1}, \qquad \frac{n}{n!} = \frac{1}{(n-1)!}$$

and

$$(c)_n(c+n) = (c)_{n+1} = c(c+1)_n = c(c+1)(c+2)_{n-1}$$

Multiplying these terms together yields:

$$C_n = -\frac{a(c-b)}{c(c+1)} \cdot \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!}$$

Therefore, summing over all $n \geq 1$:

$$\begin{aligned}
\sum_{n=1}^{\infty} C_n z^n &= -\frac{a(c-b)}{c(c+1)} z \sum_{n=1}^{\infty} \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!} z^{n-1} \\
&= -\frac{a(c-b)}{c(c+1)} z \sum_{m=0}^{\infty} \frac{(a+1)_m (b+1)_m}{(c+2)_m m!} z^m \\
&= -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1, b+1; c+2; z)
\end{aligned}$$

which establishes identity (7).

To prove identity (8), we use the symmetry of the hypergeometric function with respect to its first two parameters:

$$f(a, b+1; c+1; z) = f(b+1, a; c+1; z)$$

Applying identity (7) with parameters $(A, B, C) = (b+1, a, c+1)$, we obtain:

$$f(b+1,a;c+1;z) - f(b+1,a+1;c+2;z) = -\frac{A(C-B)}{C(C+1)} z \cdot f(A+1,B+1;C+2;z)$$
$$= -\frac{(b+1)(c+1-a)}{(c+1)(c+2)} z \cdot f(b+2,a+1;c+3;z)$$

Using symmetry once more on the functions, $f(b+1,a+1;c+2;z) = f(a+1,b+1;c+2;z)$ and $f(b+2,a+1;c+3;z) = f(a+1,b+2;c+3;z)$, which directly produces identity (8). □

## 9.2 Derivation of Gauss's Continued Fraction

We now assemble the contiguous relations into an alternating sequence of hypergeometric functions. Define the sequence of functions $\{f_n(z)\}_{n=0}^{\infty}$ by:

$$f_{2k}(z) = {}_2F_1(a+k, b+k; c+2k; z) \qquad (k \geq 0)$$
$$f_{2k+1}(z) = {}_2F_1(a+k, b+k+1; c+2k+1; z) \qquad (k \geq 0)$$

Thus the first few terms of the sequence are:

$$f_0(z) = {}_2F_1(a, b; c; z)$$
$$f_1(z) = {}_2F_1(a, b+1; c+1; z)$$
$$f_2(z) = {}_2F_1(a+1, b+1; c+2; z)$$
$$f_3(z) = {}_2F_1(a+1, b+2; c+3; z)$$
$$f_4(z) = {}_2F_1(a+2, b+2; c+4; z)$$

Observe that applying Lemma 9.2 to $(a+k, b+k, c+2k)$ gives:

$$f_{2k}(z) - f_{2k+1}(z) = -d_{2k+1} z \cdot f_{2k+2}(z)$$

where

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$

Similarly, applying identity (8) with shifted parameters gives:

$$f_{2k-1}(z) - f_{2k}(z) = -d_{2k} z \cdot f_{2k+1}(z)$$

where

$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

In general, for all $n \geq 0$, the sequence satisfies the uniform three-term recurrence relation:

$$f_n(z) - f_{n+1}(z) = -d_{n+1} z \cdot f_{n+2}(z) \tag{9}$$

**Theorem 9.3** (Gauss's Continued Fraction)**.** *Let $a, b, c \in \mathbb{C}$ with $c \notin \{0, -1, -2, \dots\}$. Then the ratio of contiguous hypergeometric functions admits the continued fraction expansion:*

$$\frac{{}_2F_1(a, b+1; c+1; z)}{{}_2F_1(a, b; c; z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{1 - \cfrac{d_3 z}{1 - \ddots}}}}$$

*where the coefficients* $\{d_n\}_{n=1}^{\infty}$ *are given by:*

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$
$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

*Proof.* Dividing the uniform three-term recurrence (9) by $f_{n+1}(z)$ (assuming $f_{n+1}(z) \neq 0$), we obtain:

$$\frac{f_n(z)}{f_{n+1}(z)} - 1 = -d_{n+1}z \cdot \frac{f_{n+2}(z)}{f_{n+1}(z)}$$

which can be rearranged as:

$$\frac{f_n(z)}{f_{n+1}(z)} = 1 - \frac{d_{n+1}z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}$$

Taking the reciprocal of both sides yields:

$$\frac{f_{n+1}(z)}{f_n(z)} = \frac{1}{1 - \dfrac{d_{n+1}z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}} \tag{10}$$

We now proceed by repeatedly substituting (10) into itself for successive values of $n$. For $n = 0$:

$$\frac{f_1(z)}{f_0(z)} = \frac{1}{1 - \dfrac{d_1 z}{\dfrac{f_1(z)}{f_2(z)}}}$$

Substituting the expression for $\frac{f_1(z)}{f_2(z)}$ from $n = 1$:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\dfrac{f_2(z)}{f_3(z)}}}}$$

Continuing this process inductively for $N$ steps:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\ddots - \cfrac{d_N z}{\dfrac{f_N(z)}{f_{N+1}(z)}}}}}$$

In the cut complex plane $\mathbb{C}\backslash[1,\infty)$, the remainder term satisfies $\lim_{N\to\infty}\frac{f_{N+1}(z)}{f_N(z)} = 1$ for $|z|<1$, ensuring convergence of the continued fraction to the meromorphic function $\frac{f_1(z)}{f_0(z)}$ [3]. Substituting the explicit formulas for $d_{2k+1}$ and $d_{2k}$ completes the proof. □

## 9.3 Special Cases and Applications

Gauss's continued fraction is remarkable because specializing the parameters $a, b, c$ and the variable $z$ yields continued fraction expansions for a large catalog of classical functions. Below we examine several key applications.

### 9.3.1 The Natural Logarithm

Recall that the Taylor series of $\ln(1+z)$ around $z=0$ is:

$$\ln(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}z^k}{k} = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots$$

Notice that the hypergeometric function with parameters $a=1, b=1, c=2$ evaluated at $-z$ is:

$$\begin{aligned} {}_2F_1(1,1;2;-z) &= \sum_{n=0}^{\infty} \frac{(1)_n(1)_n}{(2)_n n!}(-z)^n = \sum_{n=0}^{\infty} \frac{n!\,n!}{(n+1)!\,n!}(-1)^n z^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n z^n}{n+1} = \frac{\ln(1+z)}{z} \end{aligned}$$

Now set $a=1, b=0, c=1$ in Theorem 9.3. Since $(0)_0 = 1$ and $(0)_n = 0$ for $n \geq 1$, we have:

$${}_2F_1(1,0;1;-z) = 1, \qquad {}_2F_1(1,1;2;-z) = \frac{\ln(1+z)}{z}$$

Hence the ratio is:

$$\frac{{}_2F_1(1,1;2;-z)}{{}_2F_1(1,0;1;-z)} = \frac{\ln(1+z)}{z}$$

Evaluating the coefficients $d_n$ from Theorem 9.3 with $a=1, b=0, c=1$ and argument $-z$ (so $-d_n(-z) = +d_n z$):

$$\begin{aligned} d_{2k+1} &= \frac{(1+k)(1-0+k)}{(1+2k)(1+2k+1)} = \frac{(k+1)^2}{(2k+1)(2k+2)} = \frac{k+1}{2(2k+1)} \qquad (k \geq 0) \\ d_{2k} &= \frac{(0+k)(1-1+k)}{(1+2k-1)(1+2k)} = \frac{k^2}{2k(2k+1)} = \frac{k}{2(2k+1)} \qquad (k \geq 1) \end{aligned}$$

The first few coefficients are:

$$d_1 = \frac{1}{2}, \quad d_2 = \frac{1}{6}, \quad d_3 = \frac{1}{3}, \quad d_4 = \frac{1}{5}, \quad d_5 = \frac{3}{10}, \quad d_6 = \frac{3}{14}$$

Clearing the internal denominators level-by-level (by applying equivalence transformations that multiply the numerator and denominator of the $n^{\text{th}}$ level by $n$), we obtain:

**Lemma 9.4** (Gauss Continued Fraction for $\ln(1+z)$)**.** *For any $z \in \mathbb{C} \setminus (-\infty, -1]$:*

$$\ln(1+z) = \cfrac{z}{1 + \cfrac{1^2 z}{2 + \cfrac{1^2 z}{3 + \cfrac{2^2 z}{4 + \cfrac{2^2 z}{5 + \cfrac{3^2 z}{6 + \ddots}}}}}}$$

*In particular, setting $z = 1$ gives an accelerated continued fraction for $\ln(2)$:*

$$\ln(2) = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{1^2}{3 + \cfrac{2^2}{4 + \cfrac{2^2}{5 + \cfrac{3^2}{6 + \ddots}}}}}}$$

**Remark 9.1.** *Compare this expansion for $\ln(2)$ with the one obtained in Section 4.1, Lemma 4.5:*

$$\ln(2) = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

*While the expansion in Lemma 4.5 was derived from the slowly alternating harmonic series $\sum \frac{(-1)^{k+1}}{k}$ and has linear partial denominators $1$, Gauss's expansion features growing partial denominators $1, 2, 3, 4, 5, \ldots$, leading to dramatically faster numerical convergence.*

### 9.3.2 The Arctangent Function and $\pi$

Consider the series expansion of $\arctan(z)$:

$$\arctan(z) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2k+1} = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \ldots$$

Dividing by $z$:

$$\frac{\arctan(z)}{z} = \sum_{k=0}^{\infty} \frac{(-1)^k (z^2)^k}{2k+1} = \sum_{k=0}^{\infty} \frac{(1/2)_k (1)_k}{(3/2)_k k!} (-z^2)^k = {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right)$$

We set $a = 1/2, b = 0, c = 1/2$ and replace $z$ with $-z^2$ in Theorem 9.3. Then:

$$ {}_2F_1\left(\frac{1}{2}, 0; \frac{1}{2}; -z^2\right) = 1, \qquad {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right) = \frac{\arctan(z)}{z} $$

The coefficients $d_n$ are given by:

$$ d_{2k+1} = \frac{(1/2+k)(1/2-0+k)}{(1/2+2k)(1/2+2k+1)} = \frac{(2k+1)^2}{(4k+1)(4k+3)} \qquad (k \geq 0) $$

$$ d_{2k} = \frac{(0+k)(1/2-1/2+k)}{(1/2+2k-1)(1/2+2k)} = \frac{(2k)^2}{(4k-1)(4k+1)} \qquad (k \geq 1) $$

Clearing internal denominators yields:

**Lemma 9.5** (Gauss Continued Fraction for $\arctan(z)$)**.** *For $z \in \mathbb{C}\backslash\{iy : |y| \geq 1\}$:*

$$ \arctan(z) = \cfrac{z}{1 + \cfrac{1^2 z^2}{3 + \cfrac{2^2 z^2}{5 + \cfrac{3^2 z^2}{7 + \cfrac{4^2 z^2}{9 + \ddots}}}}} $$

*Evaluating at $z = 1$, since $\arctan(1) = \frac{\pi}{4}$, we obtain:*

$$ \frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{3 + \cfrac{2^2}{5 + \cfrac{3^2}{7 + \cfrac{4^2}{9 + \ddots}}}}} $$

**Remark 9.2.** *This formula for $\pi/4$ offers an instructive contrast with the continued fraction obtained in Lemma 4.7:*

$$ \frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}} $$

*In Gauss's formula, the partial numerators are the squares of consecutive integers $1^2, 2^2, 3^2, 4^2, \ldots$ and the partial denominators are the odd integers $3, 5, 7, 9, \ldots$, whereas the alternating series conversion of Section 4.2 features the odd squares in the numerators and constant 2 in the denominators.*

### 9.3.3 Confluent Limits: The Exponential and Tangent Functions

Gauss's continued fraction also extends to confluent hypergeometric functions by taking suitable limits as parameters grow without bound. Replacing $z$ by $z/b$ and taking the limit as $b \to \infty$, the hypergeometric function ${}_2F_1(a, b; c; z/b)$ tends to Kummer's confluent hypergeometric function ${}_1F_1(a; c; z)$. Applying this limit to Gauss's continued fraction produces corresponding continued fractions for ${}_1F_1(a; c; z)$.

An immediate and famous consequence is Lambert's continued fraction for the tangent function. Using the relation $\tan(z) = \frac{1}{i}\tanh(iz)$ and the continued fraction for the ratio of confluent hypergeometric functions (or directly from the hypergeometric form of $\tan(z)$), one arrives at:

**Lemma 9.6** (Lambert's Continued Fraction for $\tan(z)$)**.** *For any $z \in \mathbb{C}$ where $\tan(z)$ is defined:*

$$\tan(z) = \cfrac{z}{1 - \cfrac{z^2}{3 - \cfrac{z^2}{5 - \cfrac{z^2}{7 - \ddots}}}}$$

*Equivalently, for the hyperbolic tangent:*

$$\tanh(z) = \cfrac{z}{1 + \cfrac{z^2}{3 + \cfrac{z^2}{5 + \cfrac{z^2}{7 + \ddots}}}}$$

**Remark 9.3.** *As mentioned in Section 1, Johann Heinrich Lambert used this very continued fraction in 1761 to provide the first rigorous proof that $\pi$ is irrational. Lambert proved that if $x$ is a non-zero rational number, then $\tan(x)$ cannot be rational. Since $\tan(\pi/4) = 1$ is rational, $\pi/4$ (and hence $\pi$) must be irrational. Gauss's continued fraction provides the unified analytic framework from which Lambert's continued fraction naturally descends.*

# 10 General continued fraction examples

General continued fractions (Definition 2.2), in which the partial numerators $b_n$ are not restricted to 1, provide expressions of extraordinary elegance, symmetry, and rapid convergence for fundamental mathematical constants and functions. In this section, we develop the fundamental tool of equivalence transformations and explore celebrated general continued fractions for $\pi$, Euler's number $e$, and the Rogers-Ramanujan continued fraction.

## 10.1 Equivalence Transformations

Two continued fractions that possess identical sequences of convergents are called equivalent. Equivalence transformations allow us to rescale the partial

numerators and denominators without altering the value of any convergent, providing the principal mechanism to convert between standard and general continued fractions or to clear internal denominators.

**Definition 10.1** (Equivalence of Continued Fractions)**.** *Two continued fractions:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \ddots}} \qquad \text{and} \qquad a'_0 + \cfrac{b'_1}{a'_1 + \cfrac{b'_2}{a'_2 + \ddots}}$$

*with convergents $p_n/q_n$ and $p'_n/q'_n$ respectively, are said to be* ***equivalent*** *if:*

$$\frac{p_n}{q_n} = \frac{p'_n}{q'_n} \qquad \text{for all } n \geq 0$$

**Theorem 10.2** (Equivalence Transformation Theorem)**.** *Let $\{c_n\}_{n=1}^{\infty}$ be any sequence of non-zero constants in $\mathbb{C} \setminus \{0\}$, and set $c_0 = 1$. Then the continued fraction:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}$$

*is equivalent to the transformed continued fraction:*

$$a_0 + \cfrac{c_1 b_1}{c_1 a_1 + \cfrac{c_1 c_2 b_2}{c_2 a_2 + \cfrac{c_2 c_3 b_3}{c_3 a_3 + \ddots}}}$$

*That is, defining $a'_0 = a_0$, $a'_n = c_n a_n$, $b'_1 = c_1 b_1$, and $b'_n = c_{n-1} c_n b_n$ for $n \geq 2$, their convergents satisfy:*

$$p'_n = (c_1 c_2 \cdots c_n) p_n, \qquad q'_n = (c_1 c_2 \cdots c_n) q_n \qquad (n \geq 0)$$

*Proof.* We proceed by induction on $n$. Recall the recurrence relations for general continued fractions:

$$\begin{aligned} p_{-1} &= 1, & p_0 &= a_0, & p_n &= a_n p_{n-1} + b_n p_{n-2} \\ q_{-1} &= 0, & q_0 &= 1, & q_n &= a_n q_{n-1} + b_n q_{n-2} \end{aligned}$$

For the transformed system:

$$\begin{aligned} p'_{-1} &= 1, & p'_0 &= a'_0 = a_0 = p_0, & p'_n &= a'_n p'_{n-1} + b'_n p'_{n-2} \\ q'_{-1} &= 0, & q'_0 &= 1 = q_0, & q'_n &= a'_n q'_{n-1} + b'_n q'_{n-2} \end{aligned}$$

**Base Cases:** For $n = 0$:

$$p'_0 = a_0 = p_0, \qquad q'_0 = 1 = q_0$$

Since $c_0 = 1$, the statement holds for $n = 0$. For $n = 1$:

$$p'_1 = a'_1 p'_0 + b'_1 p'_{-1} = (c_1 a_1)a_0 + (c_1 b_1) \cdot 1 = c_1(a_1 a_0 + b_1) = c_1 p_1$$
$$q'_1 = a'_1 q'_0 + b'_1 q'_{-1} = (c_1 a_1) \cdot 1 + (c_1 b_1) \cdot 0 = c_1 a_1 = c_1 q_1$$

Thus $\frac{p'_1}{q'_1} = \frac{c_1 p_1}{c_1 q_1} = \frac{p_1}{q_1}$, confirming $n = 1$.

**Inductive Hypothesis:** Assume the formula holds for all indices up to $k - 1$, so that:

$$p'_{k-1} = (c_1 \cdots c_{k-1})p_{k-1}, \qquad p'_{k-2} = (c_1 \cdots c_{k-2})p_{k-2}$$

and similarly for $q'_{k-1}$ and $q'_{k-2}$.

**Inductive Step:** We evaluate $p'_k$ using the recurrence relation:

$$\begin{aligned} p'_k &= a'_k p'_{k-1} + b'_k p'_{k-2} \\ &= (c_k a_k) \cdot [(c_1 \cdots c_{k-1})p_{k-1}] + (c_{k-1} c_k b_k) \cdot [(c_1 \cdots c_{k-2})p_{k-2}] \\ &= (c_1 \cdots c_k)a_k p_{k-1} + (c_1 \cdots c_{k-2} c_{k-1} c_k) b_k p_{k-2} \\ &= (c_1 \cdots c_k)[a_k p_{k-1} + b_k p_{k-2}] \\ &= (c_1 \cdots c_k)p_k \end{aligned}$$

By identical algebra, the same relationship holds for $q'_k$:

$$q'_k = (c_1 \cdots c_k)q_k$$

Since each $c_j \neq 0$, the product $c_1 \cdots c_k \neq 0$. Dividing numerator by denominator:

$$\frac{p'_k}{q'_k} = \frac{(c_1 \cdots c_k)p_k}{(c_1 \cdots c_k)q_k} = \frac{p_k}{q_k}$$

This proves that all convergents coincide, establishing equivalence. □

## 10.2 General Continued Fractions for $\pi$

### 10.2.1 Lord Brouncker's Continued Fraction

The first infinite continued fraction in the history of mathematics was discovered by William, Lord Brouncker (1620–1684), first President of the Royal Society. Brouncker communicated his formula without proof to John Wallis, who published it in his *Arithmetica Infinitorum* (1656).

**Lemma 10.3** (Lord Brouncker's Continued Fraction for $\pi$)**.**

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}$$

*Equivalently, expressing $\pi$ directly:*

$$\pi = \cfrac{4}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}}$$

*Proof.* In Lemma 4.7, applying the series-to-continued-fraction theorem (Theorem 4.1) to the Leibniz series $\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1}$, we established that:

$$\frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

Taking the reciprocal of both sides directly gives:

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}$$

which is Brouncker's famous formula. □

### 10.2.2 Stern's Continued Fraction for $\pi$

In 1860, Moritz Abraham Stern published an elegant general continued fraction for $\pi$ whose partial denominators are constant:

**Lemma 10.4** (Stern's Continued Fraction for $\pi$)**.**

$$\pi = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \cfrac{7^2}{6 + \ddots}}}}$$

*Proof.* Consider the general continued fraction function:

$$f(x) = \cfrac{1^2}{2x + \cfrac{3^2}{2x + \cfrac{5^2}{2x + \ddots}}}$$

For $x = 1$, by Brouncker's formula (Lemma 10.3), $1 + f(1) = \frac{4}{\pi}$, so $f(1) = \frac{4-\pi}{\pi}$. More generally, Euler established that $f(x)$ satisfies the functional relation:

$$f(x) + x = \frac{x^2}{x + f(x+1)}$$

Evaluating at $x = 3$ gives $2x = 6$, and yields the value $f(3) = \pi - 3$. Adding 3 to both sides produces:

$$\pi = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \ddots}}}$$

□

Computing the first few convergents of Stern's continued fraction:

$$\begin{aligned}
\frac{p_0}{q_0} &= 3 = 3.000000 \\
\frac{p_1}{q_1} &= 3 + \frac{1}{6} = \frac{19}{6} \approx 3.166667 \\
\frac{p_2}{q_2} &= 3 + \frac{1}{6 + 9/6} = 3 + \frac{6}{45} = \frac{47}{15} \approx 3.133333 \\
\frac{p_3}{q_3} &= 3 + \frac{1}{6 + \frac{9}{6+25/6}} = \frac{947}{301} \approx 3.146179
\end{aligned}$$

which converge systematically to $\pi$.

### 10.2.3 Euler's Product-Derived Continued Fraction for $\pi/2$

By transforming Wallis's infinite product $\frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{4k^2}{4k^2-1} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots$, Leonhard Euler derived the continued fraction:

**Lemma 10.5** (Euler's Continued Fraction for $\pi/2$)**.**

$$\frac{\pi}{2} = 1 + \cfrac{1}{1 + \cfrac{1 \cdot 2}{1 + \cfrac{2 \cdot 3}{1 + \cfrac{3 \cdot 4}{1 + \cfrac{4 \cdot 5}{1 + \ddots}}}}}$$

## 10.3 General Continued Fractions for Euler's Number $e$

**Lemma 10.6** (Euler's General Continued Fraction for $e$)**.**

$$e = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$$

*Proof.* Consider the exponential series for $e - 2$:

$$e - 2 = \sum_{k=2}^{\infty} \frac{1}{k!} = \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots$$

Write this series as $\sum_{j=0}^{\infty} u_j$, where $u_0 = \frac{1}{2}$ and $u_j = \frac{1}{(j+2)!}$. The ratio of consecutive terms is:

$$\frac{u_j}{u_{j-1}} = \frac{1}{j+2}$$

Applying Euler's series-to-continued-fraction theorem (Theorem 4.2):

$$\sum_{j=0}^{\infty} u_0 u_1 \dots u_j = \cfrac{u_0}{1 - \cfrac{u_1}{1 + u_1 - \cfrac{u_2}{1 + u_2 - \ddots}}}$$

with $u_j = \frac{1}{j+2}$, and applying an equivalence transformation (Theorem 10.2) with $c_j = j + 1$ to clear the fractions, we obtain:

$$e - 2 = \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$$

Adding 2 to both sides completes the proof. □

The convergents of this continued fraction:

$$\begin{aligned}
\frac{p_0}{q_0} &= 2 \\
\frac{p_1}{q_1} &= 2 + \frac{1}{1} = 3 \\
\frac{p_2}{q_2} &= 2 + \frac{1}{1 + 1/2} = \frac{8}{3} \approx 2.666667 \\
\frac{p_3}{q_3} &= 2 + \frac{1}{1 + \frac{1}{2+2/3}} = \frac{30}{11} \approx 2.727273 \\
\frac{p_4}{q_4} &= 2 + \frac{1}{1 + \frac{1}{2 + \frac{2}{3+3/4}}} = \frac{144}{53} \approx 2.716981
\end{aligned}$$

converge rapidly to $e \approx 2.718281828\ldots$.

Furthermore, generalizing to the exponential function $e^x$, Euler derived the continued fraction:

$$e^x = 1 + \cfrac{x}{1 - \cfrac{x}{2 + x - \cfrac{2x}{3 + x - \cfrac{3x}{4 + x - \ddots}}}}$$

## 10.4 The Rogers-Ramanujan Continued Fraction

Among the most celebrated and profound continued fractions in analytic number theory and partition theory is the Rogers-Ramanujan continued fraction. Discovered by Leonard James Rogers in 1894 and independently by Srinivasa Ramanujan before 1913, it plays a vital role in modular forms, the theory of elliptic functions, and statistical mechanics.

**Definition 10.7** (Rogers-Ramanujan Continued Fraction)**.** *For $|q| < 1$, the **Rogers-Ramanujan continued fraction** $R(q)$ is defined by:*

$$R(q) = \cfrac{q^{1/5}}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cfrac{q^3}{1 + \cfrac{q^4}{1 + \ddots}}}}}$$

The continued fraction is intimately connected to the Rogers-Ramanujan identities:

$$G(q) = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(q;q)_n} = \prod_{n=1}^{\infty} \frac{1}{(1-q^{5n-1})(1-q^{5n-4})}$$

$$H(q) = \sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(q;q)_n} = \prod_{n=1}^{\infty} \frac{1}{(1-q^{5n-2})(1-q^{5n-3})}$$

where $(q;q)_n = \prod_{k=1}^{n}(1-q^k)$. Rogers and Ramanujan proved that $R(q)$ equals the ratio of these two theta functions:

$$R(q) = q^{1/5}\frac{H(q)}{G(q)} = q^{1/5}\prod_{n=1}^{\infty}\frac{(1-q^{5n-1})(1-q^{5n-4})}{(1-q^{5n-2})(1-q^{5n-3})}$$

**Theorem 10.8** (Ramanujan's Singular Evaluation at $q = e^{-2\pi}$)**.** *At the singular modulus $q = e^{-2\pi}$, the Rogers-Ramanujan continued fraction evaluates in closed form to an algebraic number expressed in terms of the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$:*

$$R(e^{-2\pi}) = \cfrac{e^{-2\pi/5}}{1+\cfrac{e^{-2\pi}}{1+\cfrac{e^{-4\pi}}{1+\cfrac{e^{-6\pi}}{1+\ddots}}}} = \sqrt{\frac{5+\sqrt{5}}{2}} - \frac{\sqrt{5}+1}{2} = \sqrt{\phi\sqrt{5}} - \phi$$

*Proof.* A complete proof relies on the modular transformation properties of the eta-quotients and Weber modular functions; see Berndt [6]. Numerically:

$$\phi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887$$

so

$$\sqrt{\phi\sqrt{5}} - \phi = \sqrt{1.6180339887 \times 2.2360679775} - 1.6180339887 \approx 0.2840790438$$

Computing the first few convergents of the continued fraction with $q = e^{-2\pi} \approx 0.0018674427$:

$$\frac{p_0}{q_0} = q^{1/5} = e^{-2\pi/5} \approx 0.284605$$

$$\frac{p_1}{q_1} = \frac{q^{1/5}}{1+q} \approx \frac{0.284605}{1.001867} \approx 0.284074$$

$$\frac{p_2}{q_2} = \cfrac{q^{1/5}}{1+\cfrac{q}{1+q^2}} \approx 0.28407904$$

which matches the algebraic value $\sqrt{\phi\sqrt{5}} - \phi$ to 8 decimal places in only two steps due to the rapid decay of $q^n = e^{-2\pi n}$. □

**Remark 10.1.** *In his historic first letter to G. H. Hardy dated 16 January 1913, Srinivasa Ramanujan enclosed this evaluation along with several others, notably:*

$$R(e^{-\pi}) = \frac{\sqrt{5}+1}{2} - \sqrt{\frac{5+\sqrt{5}}{2}} + 1$$

*Hardy famously wrote that these formulas:*

> *"...defeated me completely; I had never seen anything in the least like them before. A single look at them is enough to show that they could only be written down by a mathematician of the highest class. They must be true because, if they were not true, no one would have had the imagination to invent them."*

# References


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